

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

08

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

The table shows selected values from function f .

x	fx
-1	16
0	17
1	18
2	19

Which of the following is the best description of function f ?

- A. Decreasing linear
- B. Increasing linear
- C. Decreasing exponential
- D. Increasing exponential

An object travels at a constant speed of 12 centimeters per second. At this speed, what is the time, in seconds, that it would take for the object to travel 108 centimeters?

- A. 9
- B. 96
- C. 120
- D. 972

A waiter receives tips from each customer. On average, the tip is 15% of the customer's bill. At this rate, which of the following is closest to the tip the waiter can expect when a customer has a bill that is \$78.20?

- A. \$8.00
- B. \$10.00
- C. \$12.00
- D. \$14.00

Customer Purchases at a Gas Station

	Beverage purchased	Beverage not purchased	Total
Gasoline purchased	60	25	85
Gasoline not purchased	35	15	50
Total	90	40	135

On Tuesday, a local gas station had 135 customers. The table above summarizes whether or not the customers on Tuesday purchased gasoline, a beverage, both, or neither. Based on the data in the table, what is the probability that a gas station customer selected at random on that day did not purchase gasoline?

- A. $\frac{15}{50}$
- B. $\frac{15}{40}$
- C. $\frac{35}{50}$
- D. $\frac{50}{135}$

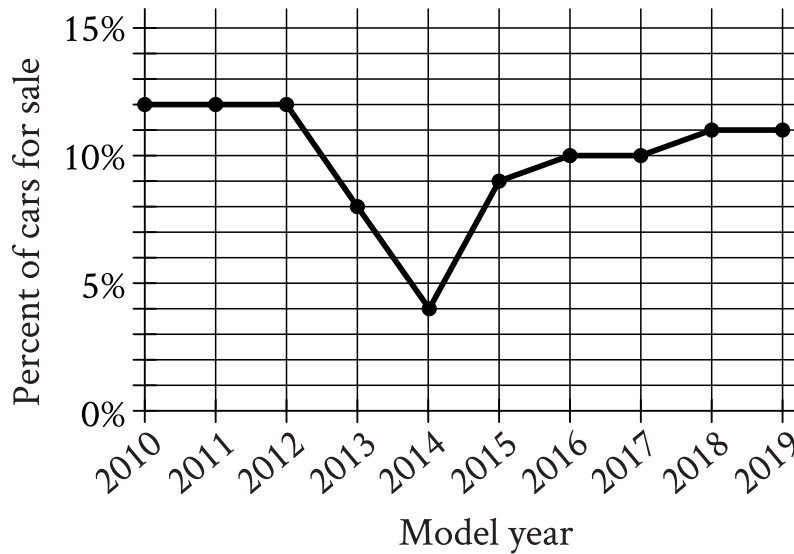
Data set A: 1, 2, 3, 4, 5, 6, 7

Data set B: 1, 1, 2, 2, 3, 3, 4

Which of the following statements correctly compares the means of data set A and data set B?

- A. The mean of each data set is 2.
- B. The mean of each data set is 4.
- C. The mean of data set A is less than the mean of data set B.
- D. The mean of data set A is greater than the mean of data set B.

The line graph shows the percent of cars for sale at a used car lot on a given day by model year.



- The line graph:
 - Begins at 2010, 12%
 - Remains level to 2011, 12%
 - Remains level to 2012, 12%
 - Falls sharply to 2013, 8%
 - Falls sharply to 2014, 4%
 - Rises sharply to 2015, 9%
 - Rises gradually to 2016, 10%
 - Remains level to 2017, 10%
 - Rises gradually to 2018, 11%
 - Remains level to 2019, 11%

For what model year is the percent of cars for sale the smallest?

A. 2012

B. 2013

C. 2014

D. 2015

An object's speed is 64 yards per second. What is the object's speed, in feet per second? 1 yard = 3 feet

- A. 61
- B. 67
- C. 94
- D. 192

Call Ratings					
	1 Star	2 Stars	3 Stars	4 Stars	Total
Employee A	16	49	72	8	145
Employee B	4	10	22	34	70
Employee C	8	56	45	16	125
Employee D	22	42	84	12	160
Total	50	157	223	70	500

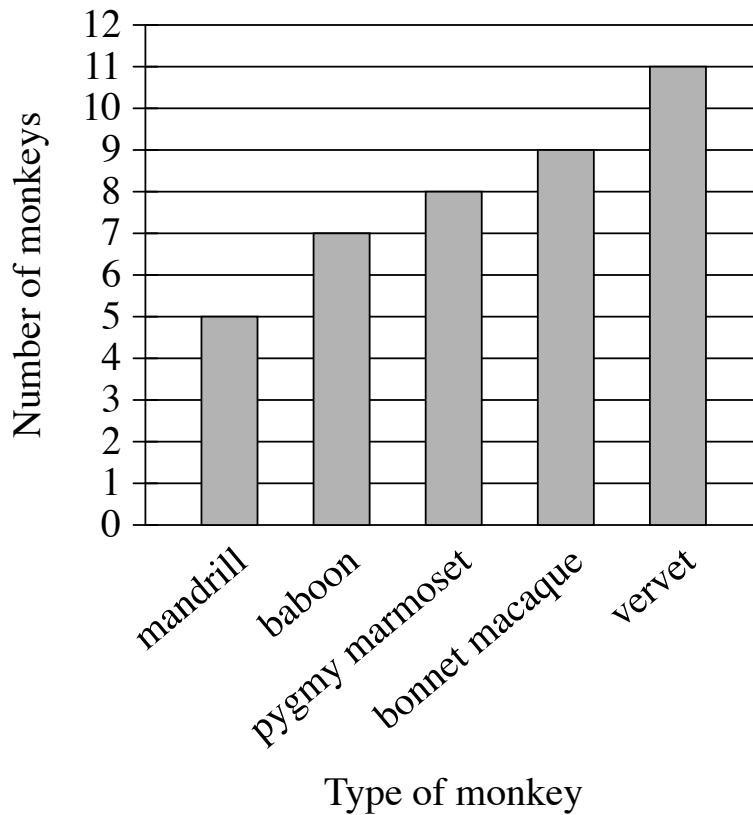
A supervisor at a call center reviewed 500 calls taken by four employees and rated the employees' performance on each call on a scale from 1 star to 4 stars. The ratings for each employee are shown in the table above. According to the table, to the nearest whole percent, what percent of Employee A's calls received a rating of 1 star?

- A. 3%
- B. 11%
- C. 16%
- D. 32%

A bag contains a total of 60 marbles. A marble is to be chosen at random from the bag. If the probability that a blue marble will be chosen is 0.35, how many marbles in the bag are blue?

- A. 21
- B. 25
- C. 35
- D. 39

The bar graph shows the number of each type of monkey in a sanctuary.



- The Number of monkeys data for the 5 bars are as follows:
 - mandrill: 5
 - baboon: 7
 - pygmy marmoset: 8
 - bonnet macaque: 9
 - vervet: 11

How many more vervets are in this sanctuary than mandrills?

A. 11

B. 6

C. 5

D. 3